The Plot Thickens

So much happened in the '70s that if we had to choose a "winner decade" from amongst the five we're talking about, this would have to be it

We promised this wouldn't be a history lesson—and it won't be. So we won't talk about the development of ARPANET and other experimental networks such as ALOHAnet—we'll make do by saying that most of the major foundations for the Internet were laid down in this decade: proposals, papers, protocols, and what not. It all developed at a mind-boggling pace—ARPANET was commissioned by the Department of Defense in 1969 to further research into networks, and in 1984, the Domain Name System was already in place! That's about 15 years from "ancient" to "modern"—and in another 15 years, the Internet was pretty much what it is now.

Networks, disks, processors, printers, programming, phones—you name it—it all happened in this decade! Naturally, with the components being developed, the personal computer—essentially a convergence of all these—had to emerge. So what enabled the quick transition from computers such as the PDP-1 to the PC as we know it? Microprocessors and disks come to mind first.

1971 saw the development of Intel's 4-bit 4004, which was quaintly dubbed "a computer on a chip" at the time. Different sources state different things about what was the first this and the first that, and you'll all too often come across the phrase "widely considered"! And so, well, The Intel 4004 is *widely considered* the world's first "commercial single-chip microprocessor." It was designed to be just a component of a calculator, but people figured it could be used for many other things. For example, it could replace certain logic chips. As more and more uses were found for the 4004, people realised that the microprocessor had tremendous potential. A trend began—that of developing more and more sophisticated microprocessors; there we see the beginnings of today's zillion-dollar microprocessor industry. However, it wasn't the 4004 that started the microcomputer revolution: the distinction is "widely considered" to belong to the 8080.

Also at the beginning of the decade, IBM introduced a floppy disk drive called the 23FD. It used a read-only 8-inch plastic disk, which held 0.816 MB. It was the first flexible disk drive—no, not "widely considered": it really was! The 5¼" flexible disk drive and diskette were introduced later, in 1976. You can imagine the impact of portable disks at a time when there were no LANs, and in 1973 came the Winchester hard disk, also by IBM.

Most hard disks today are basically Winchester disks: the Winchester was the first disk to use a "sealed head/disk assembly," which is what we use today. This wasn't the first hard disk, though: the 5 MB IBM 350 RAMAC of 1956 was the first *commercial* hard disk. And to clear the air a bit, *the* first hard disk is *widely considered* to be the IBM 350 Disk File of 1955.

AI wasn't as much in vogue as in the '60s. In fact, some felt they'd been wasting their time on the whole thing. Still, *some* things were happening: a robot called Shakey was the first to use AI for navigation. PROLOG—an important language for AI programming, more in vogue in Europe than in the US—was developed. In 1974, the first computer chess tournament was held in Stockholm. And in 1979, backgammon player Luigi Villa—then the world champion—officially became the first champion of a board game to be beaten by a computer.

Back to networks: Ray Tomlinson, the man who decided upon the "@" sign for e-mail addresses, sent the first *network* e-mail in 1971. As Tomlinson puts it, "'Network' should be included because there were many earlier instances of e-mail within a single machine." We can't figure why, though! A year later, the first computer-to-computer chat took place over ARPANET. It was between two AI programs, "Parry" and "Doctor." Parry was psychotic, and Doctor was, well, a Doctor—we have no details on whether Parry was restored to good health.

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